IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-291

1. Topic (12 words max)

Real-Time Driver and Road Monitoring for Risk Prediction

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	25-26J-291
Year	2025

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Road accidents remain one of the leading causes of injury and death globally, with developing countries like Sri Lanka facing unique challenges due to poor infrastructure, unpredictable driver behavior, and limited access to emergency response systems. Despite advances in intelligent transport systems (ITS), most solutions are either cloud-dependent, fragmented in scope, or too expensive for wide deployment in low-resource settings. This research addresses the need for an affordable, integrated, and real-time accident prevention and response system tailored to Sri Lankan conditions.

The core objective is to develop a **Real-Time Risk Prediction and Emergency Response System** powered by **Machine Learning (ML)** deployed on a microcontroller. The system fuses data from multiple modules—**Driver Condition Monitoring**, **Driving Pattern Analysis**, and **Road Condition Monitoring**—to assess accident risk in real time and trigger alerts or emergency responses when needed. This approach enables local, offline, and instant decision-making using low-cost hardware such as the ESP32,Raspberry PI or similar embedded systems.

Each module plays a vital role: driver condition monitoring detects drowsiness or distraction using computer vision; driving pattern analysis captures erratic behavior such as harsh braking or overspeeding using IMU sensors; and road condition monitoring identifies hazards like potholes or uneven surfaces. All data streams are processed through an ML model trained on real or simulated local datasets, ensuring accuracy and relevance.

By integrating all components into a unified, edge-based system, this research aims to create a practical solution for accident prediction and emergency management in real-world conditions. It will also contribute to academic understanding in embedded ML, sensor fusion, and risk modeling, with potential for real deployment in vehicles and road safety initiatives across Sri Lanka.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

This research proposes a real-time, edge-based accident prevention and emergency response system that leverages machine learning (ML) and multi-sensor data fusion. The system is designed to function independently of cloud connectivity, enabling real-time decision-making and emergency response in both urban and rural areas of Sri Lanka.

The solution integrates four key modules:

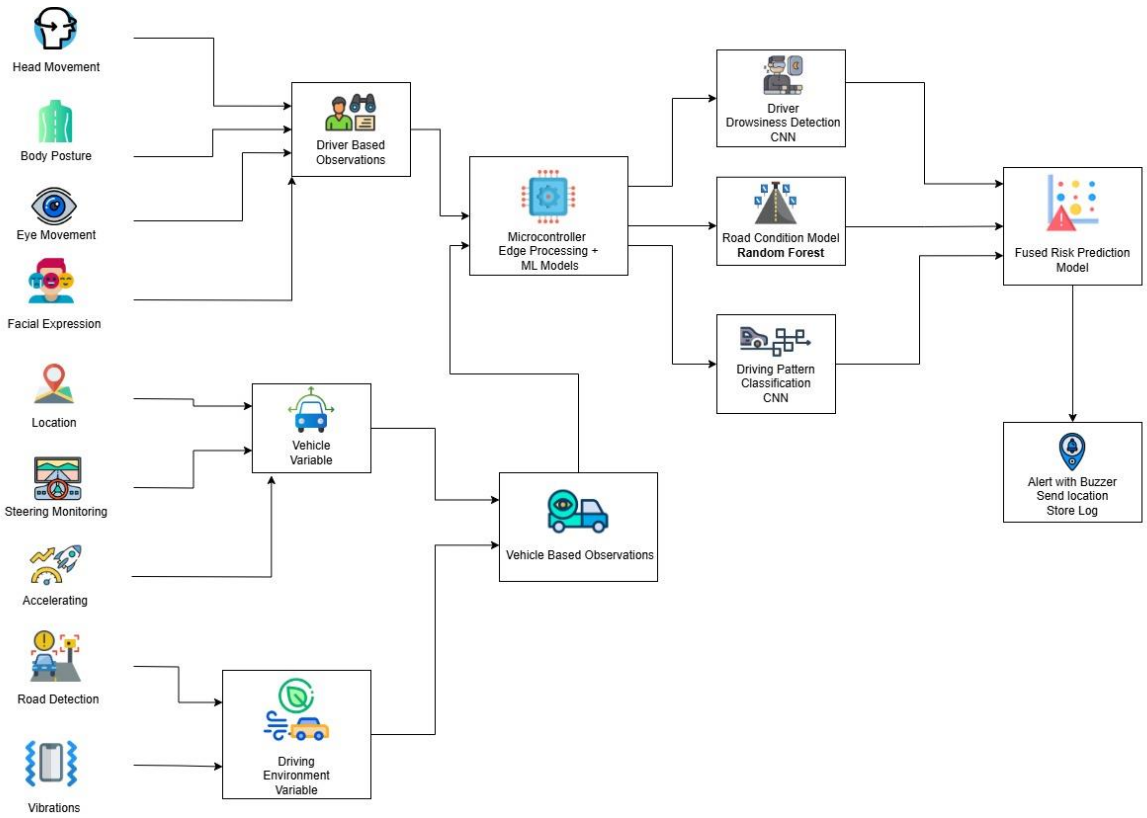
1. Driver Condition Monitoring – uses image processing (e.g., facial landmarks or IR sensing) to detect drowsiness, distraction, or fatigue.

2. Driving Pattern Analysis – collects and analyzes data from IMU (accelerometer and gyroscope) sensors to recognize abnormal driving behavior like harsh braking, overspeeding, or erratic lane changes.

3. Road Condition Monitoring – uses vibration sensors and cameras to detect potholes, slippery roads, or surface irregularities that increase accident risk.

4. Real-Time Risk Prediction & Emergency Response – an ML model deployed on a microcontroller (e.g., ESP32) processes real-time data from all modules to predict the likelihood of an accident. If risk is high, the system triggers a driver alert and automatically sends an emergency SMS with GPS location.

The innovation lies in running the ML model directly on a low-power microcontroller, ensuring instant response, energy efficiency, and affordability. This integrated system is specifically tailored to the Sri Lankan context, considering common road hazards, driving behavior, and emergency accessibility challenges.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This research operates at the intersection of Machine Learning (ML), Embedded Systems, Intelligent Transportation Systems (ITS), and Sensor Integration. It requires domain expertise in real-time sensor data processing, TinyML and TensorFlowLite model development, and vehicle behaviour analysis, specifically tailored for Sri Lanka. From the ML perspective, the project involves collecting and preprocessing time-series and image data from accelerometers, gyroscopes, cameras, and vibration sensors. The models must be lightweight (quantized neural networks, decision trees) and suitable for deployment on microcontrollers such as the ESP32, Raspberry PI, making TinyML and TensorFlowLite knowledge essential. Understanding model compression, latency management, and embedded deployment workflows is key. In terms of sensor and embedded systems, expertise is needed in configuring and

calibrating sensors like MPU6050 (IMU), IR eye blink sensors, vibration sensors, and GSM/GPS modules. The team must also handle real-time communication, power optimization, and hardware-software integration. For data requirements, the project needs labeled datasets for:

• Drowsiness and driver distraction detection (YawDD, EyeBlink8)
https://www.kaggle.com/datasets/ismailnasri20/driver-drowsiness-dataset-ddd

• Driving behavior patterns (UAH-DriveSet, or custom IMU data)
https://ieee-dataport.org/documents/aggressive-driving-behavior-iot-data?utm_source=chatgpt.com

• Road condition classification (pothole or bump datasets like RDD2020 or locally collected signals)
https://drive.google.com/drive/folders/1KHTeyLbqYgnQXEgTeOWsB6Z7wOK7usue

• Accident scenarios for training the risk prediction model (synthetic fusion or custom labeled logs)

Since many global datasets may not reflect local conditions, the team plans to collect contextual data from real Sri Lankan environments to enhance model relevance. Overall, this project demands a multidisciplinary skill set combining ML engineering, embedded hardware design, and domain-specific transportation knowledge to build a scalable and intelligent accident prevention system.

8. Objectives and Novelty

Main Objective To design and implement a **real-time, microcontroller-based accident risk prediction and emergency response system** by integrating data from driver condition, driving patterns, and road surface monitoring — optimized for Sri Lankan road environments.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Driver Condition Monitoring	Detect drowsiness or distraction using real-time visual or sensor-based input.	➢ Use camera or IR sensor to monitor driver's eye blink and head position. ➢ Preprocess face data and detect fatigue signs. ➢ Deploy lightweight detection models for microcontroller/embedded use.	➢ Low-cost, camera-based drowsiness detection designed for local lighting and road conditions. ➢ Runs without internet/cloud — embedded processing on edge device. ➢ Tailored for conditions unique to Sri Lankan drivers (e.g., heat, glare, dusty cabins).
Driving Pattern Analysis	Analyze risky driving behaviors using IMU sensor data.	➢ Collect accelerometer and gyroscope data during real driving.	➢ Real-time behavioral classification on a microcontroller using IMU data.

		➢ Identify unsafe patterns like harsh braking, swerving, and overspending. ➢ Train Tiny ML models for real-time classification.	➢ Adapted to driving irregularities specific to Sri Lanka. ➢ Enables early detection of risky patterns even in low-tech vehicles.
Road Condition Monitoring	Detect potholes, bumps, and poor road conditions via vibration or camera sensors.	➢ Mount vibration sensors or bottom-facing cameras. ➢ Label and classify signals using signal processing or ML. ➢ Integrate GPS tagging for hazardous location mapping.	➢ Offline, sensor-based alternative to expensive mobile or cloud systems. ➢ Designed to work on rough or rural Sri Lankan roads. ➢ Enables localized road condition mapping for safety and infrastructure improvement.
Real-Time Risk Prediction & Emergency Response (ML on Microcontroller)	Fuse data from all modules to predict accident risk and send emergency alerts.	➢ Design risk scoring model using inputs from Modules 1–3. ➢ Train and quantize ML model for ESP32 or Raspberry PI deployment. ➢ Trigger emergency alerts (buzzer, LED, SMS) with GPS coordinates.	➢ First-time integration of driver, vehicle, and road inputs on an embedded ML system for accident prevention. ➢ Works offline — ideal for remote regions with poor connectivity. ➢ Emergency response system tailored for real-world constraints in Sri Lanka

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
DILOCHANA D L R IT22211514	**Real-Time Risk Prediction & Emergency Response (ML on Microcontroller)** **Description:** This is the integrative module that collects data from other modules, fuses it to assess real-time risk, and triggers emergency actions if needed. **IT Relevance:** • Involves embedded system programming (Raspberry Pi) • Uses ML model deployment frameworks (TensorFlow Lite) • Implements event-based systems and communication protocols (GPS/GSM) • Aligns with IT system design, integration, and real-time processing
SENADHEERA S C C IT22134226	**Driving Pattern Analysis** **Description:** Analyzes vehicle behavior using IMU sensors, GPS, and accelerometer to classify driving behavior (rash, normal, distracted). **IT Relevance:** • Use of ML models (Random Forest, LSTM) to classify time-series data • Sensor data handling and preprocessing • Data analytics and behavior prediction • Deployment on edge devices (embedded IT systems)

SAMODA T W O IT22257468	**Road Condition Monitoring** **Description:** Detects road anomalies like potholes, bumps, or poor surfaces using vibration sensors or cameras with ML. **IT Relevance:** • IoT integration (vibration sensors, cameras) • Applies ML for image/sensor classification • Use of data acquisition, processing, and visualization • Promotes smart infrastructure **concepts** via IT
PINSARA T H A K IT22255860	**Driver Condition Monitoring** **Description:** This module uses a camera and computer vision algorithms to monitor facial features (eyes, head pose, expressions) to detect drowsiness or inattentive behavior. **IT Relevance:** • Applies Computer Vision (OpenCV, MediaPipe) • Utilizes ML algorithms (CNN, SVM) • Involves real-time image processing • Knowledge of software-hardware interfacing (Raspberry Pi with camera)

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mrs.	Manori	Gamage	23/06/2025.
Co-Supervisor				

External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Recomended.

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.